

# Decoupling Analog Circuit Representation from Technology for Behavior-Centric Optimization

Jintao Li<sup>1,2,†</sup>, Haochang Zhi<sup>3,†</sup>, Jiang Xiao<sup>1</sup>, Keren Zhu<sup>2,\*</sup>, Yun Li<sup>1,\*</sup>

<sup>1</sup>Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen 518110, China

<sup>2</sup>School of Microelectronics, Fudan University, Shanghai 200433, China

<sup>3</sup>School of Electronic Science and Engineering, Southeast University, Nanjing 214135, China

**Abstract**—Analog IC design is mainly manual and implemented at the device level. A major reason is circuit behavior-extraction. Unlike its digital counterpart, analog IC design is strongly coupled with technology nodes and is difficult to represent by an abstract behavioral model. The lack of accurate and efficient analog modeling has become a bottleneck in analog design automation. This paper proposes a behavior-centric optimization framework for analog circuits that represents circuit behavior using transistor electrical properties instead of sizes, improving model generalization and reducing optimization complexity. To characterize the process, we propose a method for mapping transistor electrical properties to sizes. Moreover, we developed a radial basis functions-based Kolmogorov-Arnold network (RBF-KAN) to accurately approximate circuit nonlinear behavior with limited simulations. Compared to black-box modeling, our approach enables constructing surrogate models via KAN under a set specification with just a few hundred simulations. Experiments on the testing suite showed our framework achieved a  $1.76\times$  to  $2.64\times$  improvement in large signal figure of merit (FOM) and  $1.73\times$  to  $2.48\times$  in small signal FOM over state-of-the-art methods, while also enabling  $3.5\times$  to  $6.2\times$  acceleration in design porting.

**Index Terms**—Analog circuit sizing, Kolmogorov-Arnold Networks, Knowledge Transfer, Behavioral Model

## I. INTRODUCTION

While digital design benefits from mature CAD tools, analog design still relies on experienced experts. However, as process technology advances and application demands increase, stricter design specifications drive the need for greater automation and intelligence in analog IC sizing [1]–[3].

Currently, most approaches rely on treating the analog sizing problem as a black box, employing machine learning (ML) techniques like reinforcement learning (RL) [4], [5], Bayesian optimization (BO) [6], and evolutionary algorithms (EA) [3], [7] to explore the design space. These methods often overlook the circuit's inherent behavior, instead relying on extensive computations to meet specifications, requiring substantial simulation time and constructing accurate surrogate models.

Considering that circuits are commonly represented as schematics, existing learning algorithms employ graph neural networks (GNNs) [8] to leverage topological information to open the black box. Nonetheless, this graph representation mainly encodes device connectivity, providing limited insight into the circuit's deeper behavior, thus still requiring **thousands of simulations** to build an accurate surrogate model.

Deriving more meaningful design insights requires establishing signal models and response analysis, which link the electrical parameters of devices to circuit performance, enabling guiding designs. In fact, early analog electronic design automation (EDA) methods employed **symbolic** approaches [10] to approximate circuit equations rather than relying solely on simulation. For human experts, a commonly used approach is the  $g_m/I_D$  methodology [9], which characterizes circuit

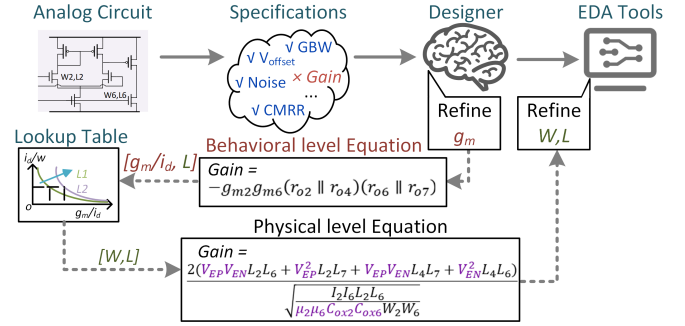


Fig. 1. Designers approximate the behavioral equation for performance metrics based on circuit theory and then utilize the  $g_m/I_D$  [9] methodology mapping to specific design parameters.

performance based on transconductance ( $g_m$ ) to drain current ( $I_D$ ). As shown in Figure 1, this approach provides behavior-level insights that reduce reliance on specific process design kits (PDKs) data, enabling more efficient optimization than simulation-based methods. While this approach was efficient, it gradually fell out of favor due to three main challenges: (1) Technology scaling introduced non-idealities, making accurate equation-based approximation difficult. (2) Fitting separate equations was required for each specification and circuit topology. (3) Mapping behavior to design parameters requires **presetting the transistor length ( $L$ )**, making the design process dependent on trial and error and limiting cross-process portability.

Revisiting these limitations of the classic equation-based methods, we observed that the key challenges lie in the following: (1) Due to the **strong coupling with the PDKs**, building surrogate models for analog blocks requires capturing their highly nonlinear properties and fitting the corresponding process characteristics. (2) ML is primarily based on **connectionism**, while circuit representation relies on **symbolism**, which hinders effective integration. (3) Expanding the objective space with complete specifications leads to **exponential** complexity in managing the Pareto front (PF).

We propose a framework for behavioral-centric optimization of analog circuits (BCOA) to address these challenges. Inspired by human expert workflows, we decouple the analog block from the process. Specifically, we approximate circuit behavior with **process-independent electrical properties** and have developed a **direct mapping method** from these properties to transistor sizes, eliminating the need to preset  $L$  and facilitating porting. This enables optimization and modeling to directly focus on analog behavior, bridging the gap between symbolic and connectionist approaches.

The contributions of this work are summarized as follows.

- 1) We decouple the analog block from the process and propose

\*Corresponding authors: (Krzhu@fudan.edu.cn, Yun.li@ieee.org).

†Both authors contributed equally to this research.

a behavioral-level optimization framework that reduces the complexity of modeling while enhancing generalization across different technologies.

- 2) We propose a method to map electrical properties to transistor sizes and develop a radial basis function-based Kolmogorov-Arnold network (RBF-KAN) model to learn circuit nonlinear behavior from limited simulations efficiently.
- 3) We propose a method for Pareto front estimation that reduces the time complexity from  $O(n^m)$  to  $O(n^2 \cdot \log n + n \cdot m)$ , enabling efficient complete specification optimization.
- 4) Experiments based on the AnalogGym suite show that the proposed framework builds accurate behavioral-level surrogate models with only a few hundred simulations and effectively generalizes across different processes. Compared to state-of-the-art algorithms, BCOA achieves a  $1.73$  to  $2.64 \times$  improvement in  $FOM_L$  and  $FOM_S$ .

## II. PRELIMINARIES

### A. Surrogate Model for Analog Blocks

Gaussian processes (GP) [6], [11] is among the most popular surrogate modeling methods due to their strong learning ability with limited simulations. However, considering complete specifications, often ranging from 10 to 20, GPs struggle with scalability, causing computational costs to grow exponentially with more objectives or samples [2].

Recent research explores to use deep learning for surrogate modeling due to its scalability for multiple objectives. However, this approach requires large amounts of data for effective training and struggles to model nonlinear transistor behavior accurately. In contrast, KAN [12] uses learnable B-splines as activation functions to decompose complex multivariate functions into simpler univariate components. We observe that this property allows KANs to efficiently capture nonlinear relationships with limited data, providing more accurate and scalable learning.

### B. Design Flow with $g_m/I_D$ Methodology

$g_m/I_D$  methodology [9], [13] is a widely-used design methodology that links small-signal models with transistor sizes. In the  $g_m/I_D$  design methodology, transistor characteristics for various channel lengths  $L$  and widths  $W$  are simulated and measured, capturing the relationship between  $g_m/I_D$  and the normalized current  $I_D/(W/L)$ . This data is then compiled into pre-simulated lookup tables (LUTs). Subsequently, the designer selects the appropriate operating region based on the performance requirements. The weak inversion region is typically chosen for low-power applications, where the  $g_m/I_D$  is higher. In contrast, the strong inversion region, characterized by a lower  $g_m/I_D$ , is preferred for high-speed applications. Once the operating area is defined, the designer utilizes the LUT to determine the appropriate device dimensions.

### C. Automatic Analog Sizing

Recent research trends on analog sizing often focus on efficiency. To enhance optimization efficiency, techniques like truncated subspace sampling [14] and domain decomposition [15] have been introduced, targeting specific areas of the design space to reduce computation and accelerate convergence. However, these methods may potentially miss points with better circuit performance. We investigate an amplifier optimization task with seven objectives conducted on the AnalogGym testing suite [16], domain decomposition-based BO reduced the required simulation runs by a factor of 10 compared to EA. However, this improvement in efficiency came at a substantial

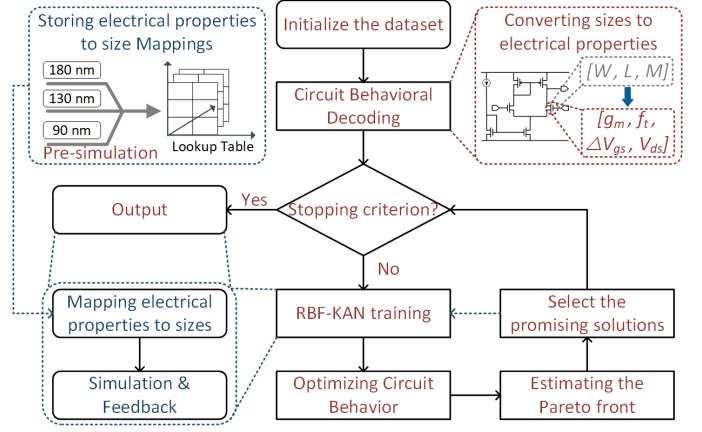


Fig. 2. The overall flow of the proposed BCOA framework.

cost: the figure of merit (FOM) dropped by **10 to 100**  $\times$ . In BCOA, we use *behavioral representation* to reduce optimization complexity and enhance model generalization, along with efficient PF estimation to guide the many-objective optimization.

## III. FRAMEWORK

### A. Framework Overview

BCOA mainly consists of six steps, as shown in Figure 2. 1) We use pre-simulation to build LUTs that map electrical properties to transistor sizes, as outlined in Section III-B. 2) We apply Latin hypercube sampling [17] to generate a small set of candidate designs as a database and use simulations to obtain transistor electrical properties that represent circuit behavior. 3) If the stopping criteria are met, the best design is selected; otherwise, we proceed with further optimization. 4) We train the RBF-KAN model using the solutions detailed in Section III-C and their corresponding performance from the database as training data. For the new solutions, we use the LUTs from Step 1 to map transistor electrical properties back to sizes and simulate to obtain the corresponding performance. 5) We apply an optimization algorithm to refine circuit behavior using predictions from the RBF-KAN. Notably, the BCOA framework is compatible with various optimization algorithms, and we select MOEA/D-DE [18] as the optimizer for its efficiency and flexibility. 6) The PF is estimated following the method in Section III-D, allowing us to select the most promising solutions for the next iteration.

### B. Transistor Size Mapping Method

We develop a transistor size mapping method to decouple circuit behavior from technology, overcoming the  $g_m/I_D$  methodology's reliance on presetting  $L$  to determine size. Therefore, the key idea of our method is to use large-signal and small-signal parameters to characterize transistor electrical properties and, building on the  $g_m/I_D$  methodology, to fix  $L$  using these parameters, as shown in Figure 3. The small-signal parameters include  $g_m$ , which represents the change in drain current due to a change in gate voltage, and transition frequency  $f_t$ , which measures high-frequency performance. These parameters determine bandwidth, slew rate (SR), power consumption, phase, and transient response. The large-signal parameters, including drain-source voltage  $V_{ds}$  and overdrive voltage  $\Delta V_{gs}$  (defined as  $V_{gs} - V_{th}$ , where  $V_{gs}$  is the gate-source voltage and  $V_{th}$  is the threshold voltage), define the transistor's operating region and influence swing, gain, and noise performance.

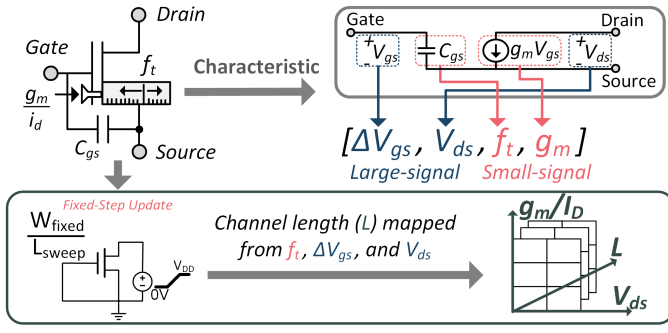


Fig. 3. We use large-signal and small-signal parameters to characterize transistors, providing a more accurate mapping from electrical properties to design parameters than the  $g_m/I_D$  methodology.

The mapping between electrical properties and transistor sizes uses the formulas in Table I to calculate  $L$ , and the corresponding sizes are then derived from the  $g_m/I_D$  LUT. Further details on constructing a LUT with the  $g_m/I_D$  method can be found in [13].

TABLE I  
CORRELATION BETWEEN ELECTRICAL PROPERTIES AND SIZES

| Parameter           | Formulas                                                | Relationship          |
|---------------------|---------------------------------------------------------|-----------------------|
| $g_m (A/V)$         | $\sqrt{2\mu C_{ox} \frac{W}{L} I_{ds}}$                 | $\sqrt{W/L}$          |
| $f_t (Hz)$          | $\frac{3\mu I_{ds}}{2\pi g_{m0} L^2}$                   | $1/L^2$               |
| $\Delta V_{gs} (V)$ | $\sqrt{\frac{4\pi f_{t0} g_{m0} L^3}{3\mu^2 C_{ox} W}}$ | $L\sqrt{\frac{L}{W}}$ |
| $V_{ds} (V)$        | Non-analytic                                            | $\sim L/W$            |

### C. RBF-KAN

We adopt RBF-KAN to enable data-efficient and accurate modeling of circuit behavior, enhancing scalability and sensitivity to both global and local features. In KAN, each node  $x_j^{l+1}$  in layer  $l+1$  is computed as the sum of activation functions  $\phi_{l,i,j}$  applied to nodes  $x_i^l$  in the previous layer:

$$x_j^{l+1} = \sum_{i=1}^{n_l} \phi_{l,i,j}(x_i^l), \quad (1)$$

where  $x_i^l$  represents the  $i$ -th node in layer  $l$ , and  $\phi_{l,i,j}$  denotes the activation function on the edge. The B-splines are adaptable basis functions for activation, enhancing nonlinear fitting capabilities. However, their grid-based structure limits flexibility with non-uniform data and increases computational complexity.

To enhance global smoothness and long-range feature capture with better scalability in high dimensions, we replace B-splines with Gauss-Laplace RBF activations [19], defined as:

$$\phi_{l,i,j}(x_i^l) = W_{ij}^l \cdot \exp\left(-\alpha(x_i^l - g_{ij}^l)^2 - \beta|x_i^l - g_{ij}^l|\right), \quad (2)$$

where  $\alpha$  controls the scale of the Gaussian component, capturing smooth variations and low-frequency features, and  $\beta$  determines the decay rate of the Laplace component, enhancing sensitivity to rapid changes and high-frequency signals.  $W_{ij}^l$  represents the edge weight,  $g_{ij}^l$  denotes the RBF center, and  $\alpha$  and  $\beta$  are hyperparameters that control the RBF shape.

For backpropagation, the gradient of the loss function to the weight  $W_{ij}^l$  is:

$$\frac{\partial L}{\partial W_{ij}^l} = \delta_j^{l+1} \cdot \phi_{l,i,j}(x_i^l), \quad (3)$$

where  $\delta_j^{l+1} = \frac{\partial L}{\partial x_j^{l+1}}$ . The gradient for the center  $g_{ij}^l$  requires the partial derivative of the activation function concerning  $g_{ij}^l$ :

$$\frac{\partial \phi_{l,i,j}}{\partial g_{ij}^l} = W_{ij}^l \cdot \phi_{l,i,j}(x_i^l) \left(2\alpha(x_i^l - g_{ij}^l) + \beta \cdot \text{sgn}(x_i^l - g_{ij}^l)\right), \quad (4)$$

which leads to the gradient as  $\frac{\partial L}{\partial g_{ij}^l} = \delta_j^{l+1} \cdot \frac{\partial \phi_{l,i,j}}{\partial g_{ij}^l}$ .

To propagate gradients to the previous layer, we compute the gradient concerning the input  $x_i^l$ :

$$\frac{\partial L}{\partial x_i^l} = \sum_{j=1}^{n_{l+1}} \delta_j^{l+1} \cdot \frac{\partial \phi_{l,i,j}}{\partial x_i^l}, \quad (5)$$

where the partial derivative of the activation function for  $x_i^l$  is:

$$\frac{\partial \phi_{l,i,j}}{\partial x_i^l} = W_{ij}^l \cdot \phi_{l,i,j}(x_i^l) \left(-2\alpha(x_i^l - g_{ij}^l) - \beta \cdot \text{sgn}(x_i^l - g_{ij}^l)\right). \quad (6)$$

### D. Pareto Front Estimation

We propose a Monte Carlo-based hypervolume (MC-HV) estimation method to enhance objective scalability and effectively guide optimization. MC-HV approximates the HV by dynamically adjusting reference points closer to the PF and increasing sampling density in complex regions according to curvature. Algorithm 1 illustrates the MC-HV procedures. The  $\epsilon$  controls the convergence precision of the reference point, typically set to  $10^{-3}$  or  $10^{-4}$ . The  $\Delta$ , defined as 10% of the objective range of the solution set, ensures that the initial reference point is sufficiently far from the solution set. The  $\beta = 0.5$  regulates the rate at which the reference point approaches the PF, while the  $\gamma = 1$  adjusts the impact of the distance term in curvature weighting. The  $\alpha = 100$  influences the total sample count, and the  $\beta = 0.5$  increases sampling density in high-curvature regions. The distance term  $d(z, S) = \min_{s \in S} \|z - s\|$  in  $p(z)$  concentrates sampling near the solution set to enhance HV estimation accuracy. In high-curvature regions, the sampling count increases adaptively, with  $N_i = \alpha \cdot (1 + \kappa(s_i))^\beta$ , ensuring more sampling points in complex regions to improve accuracy.

The time complexity of the MC-HV method primarily depends on the solution set size  $n$  and the number of objectives  $m$ . The hypervolume estimation complexity is mainly due to dominance checks on each sampling point, where each check has a complexity of  $O(n \cdot \log n)$ , resulting in a complexity of  $O(n^2 \cdot \log n)$ . The reference point adjustment complexity is  $O(n \cdot m)$ , with  $m$  representing the number of objectives.

### E. Design Porting

Behavioral-level optimization enables more efficient and flexible cross-process portability than transistor sizing. Figure 4 illustrates a case study of circuit behavior-based design porting from 180 nm to 130 nm. Since our framework represents the circuit using process-independent transistor electrical properties, the optimized circuit behavior at 180 nm is mapped to specific dimensions at 130 nm through the LUT of the 130 nm process. As process nodes shrink, both  $V_{th}$  and supply voltage ( $V_{dd}$ ) decrease to enhance speed and reduce power, so fine-tuning primarily focuses on the  $\Delta V_{gs}$  and  $V_{ds}$ . For the RBF-KAN, we implement knowledge transfer using hard parameter sharing [20].

TABLE II  
SUMMARY OF OPTIMIZATION PERFORMANCE AVERAGED OVER 10 RUNS FOR DIFFERENT TOPOLOGIES

| Topology                         | NMCf         |              |               |               | NMCNR   |              |               |               | DFCFC1 |              |               |               |
|----------------------------------|--------------|--------------|---------------|---------------|---------|--------------|---------------|---------------|--------|--------------|---------------|---------------|
| Algorithm                        | rGNNRL       | cVTSBO       | $g_m/I_{DRL}$ | BCOA          | rGNNRL  | cVTSBO       | $g_m/I_{DRL}$ | BCOA          | GCNRL  | cVTSBO       | $g_m/I_{DRL}$ | BCOA          |
| PM (°)                           | 64.3         | 59.5         | 51.2          | 49.6          | 67.2    | 60.3         | 48.2          | 52.4          | 52.7   | 59.4         | 55.4          | 49.6          |
| Gain (dB) ↑                      | 127.6        | <b>137.2</b> | 126.4         | 119           | 125.6   | 123.4        | <b>129.4</b>  | 116.4         | 119.6  | <b>127.1</b> | 125.6         | 124.1         |
| PSRR (dB) ↓                      | -73.1        | -70.3        | -70.5         | <b>-75.4</b>  | -67.5   | -58.3        | -75.2         | <b>-89.6</b>  | -74.4  | -66.2        | -72.4         | <b>-76.2</b>  |
| CMRR (dB) ↓                      | <b>-72.9</b> | -69.4        | -68.2         | -67.2         | -74.2   | -46.2        | -80.5         | <b>-88.5</b>  | -73.7  | -67.5        | -77.6         | <b>-79.4</b>  |
| $T_s$ (μs) ↓                     | 5.4          | 3.6          | 3.2           | <b>2.2</b>    | 15.2    | 14.2         | 9.6           | <b>2.3</b>    | 15.6   | 16.4         | 10.6          | <b>0.6</b>    |
| FOM <sub>L</sub> (V/μs*pF/mW) ↑  | 192.1        | 171.4        | 256.1         | <b>451.9</b>  | 152.3   | 175.4        | 219.4         | <b>396.2</b>  | 416.2  | 408.25       | 506.2         | <b>964.2</b>  |
| FOM <sub>S</sub> (MHz*pF/mW) ↑   | 436.4        | 485.9        | 624.6         | <b>1081.4</b> | 894.2   | 977.6        | 952.6         | <b>1409.4</b> | 774.2  | 767.23       | 824.5         | <b>1352.7</b> |
| Active Area (mm <sup>2</sup> ) ↓ | 142.5        | 139.7        | 158.2         | <b>121.5</b>  | 296.4   | 245.6        | 257.6         | <b>232.7</b>  | 109.5  | 84.1         | 119.6         | <b>71.6</b>   |
| FOM <sub>AMP</sub> ↑             | 19.1         | 28.5         | 49.4          | <b>283.5</b>  | 4.9     | 4.2          | 17.0          | <b>186.2</b>  | 31.8   | 33.2         | 59.8          | <b>5863.1</b> |
| FEs Consumption ↓                | 12257.2      | <b>122.4</b> | 8554.7        | 3665.2        | 10598.2 | <b>116.8</b> | 8752.4        | 2982.4        | 9624.5 | <b>120.4</b> | 10265.4       | 3106.5        |
| Simulation Count ↓               | 5274.5       | <b>122.4</b> | 3254.3        | 401.2         | 4106.4  | <b>116.8</b> | 3626.4        | 426.5         | 3962.4 | <b>120.4</b> | 4216.5        | 376.4         |
| Total Runtime (min) ↓            | 413.5        | <b>8.8</b>   | 224.3         | 22.6          | 352.6   | <b>8.9</b>   | 294.5         | 19.4          | 336.1  | <b>9.6</b>   | 354.2         | 20.2          |

#### Algorithm 1 MC-HV Method

**Input:** Solution set  $S$ , termination threshold  $\epsilon$ , sampling parameter  $\gamma$ , sampling control parameter  $\alpha$ , curvature influence parameter  $\beta$ , number of samples  $N$

- 1: Initialize  $r$  for each objective:  $r_{i,\text{initial}} = \max_{s \in S} s_i + \Delta$
- 2: **while**  $\max_i (r_i - \max_{s \in S} s_i) > \epsilon$  **do**
- 3:   Update each dimension of reference point  $r$ :  
 $r_i = \max_{s \in S} (s_i + \beta(r_{i,\text{initial}} - s_i))$
- 4: **end while**
- 5: Compute curvatures for each solution  $s_i \in S$ :  
 $\kappa(s_i) = \frac{\|\mathbf{n}(s_i) - \mathbf{n}(s_j)\|}{\|s_i - s_j\|}$
- 6: Initialize sample set:  $S \leftarrow \emptyset$
- 7: **for**  $i = 1, \dots, N$  **do**
- 8:   Sample  $z \sim p(z) = \frac{1}{Z} \exp\left(-\gamma \cdot d(z, S) \cdot \frac{1}{1+\kappa(z)}\right)$
- 9:    $S \leftarrow S \cup \{z\}$
- 10: **end for**
- 11: Estimate hypervolume  $HV(S)$  using Monte Carlo integration:  
 $HV(S) \approx \frac{1}{N} \sum_{z \in S} \frac{\mathbb{I}(z \text{ is dominated by } S)}{p(z)}$
- 12: **return**  $HV(S)$

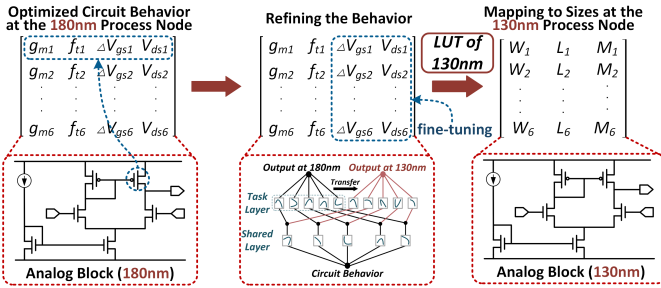


Fig. 4. Our framework enables design porting by mapping optimized behavior to transistor sizes at the new process node via an LUT, with design knowledge transferred through hard parameter sharing in the surrogate model.

## IV. EXPERIMENTAL RESULTS

We use the reference metric  $FOM_{AMP}$  provided by AnalogGym [16] to evaluate the optimization results by comparing the optimized performance metrics against the baseline metrics, which are also

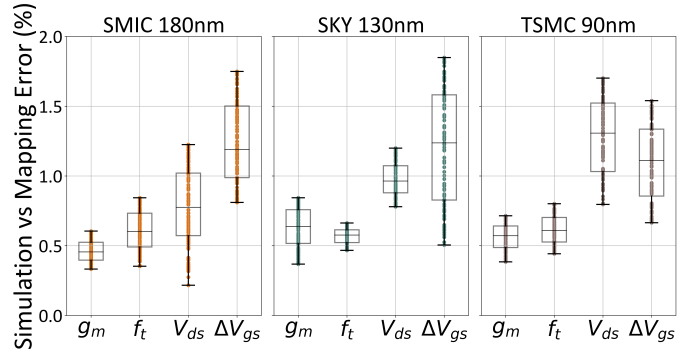


Fig. 5. Error comparison of transistor size mapping method with simulation across three process nodes. Each experiment was repeated ten times with different random seeds.

already offered within AnalogGym. The formula is as follows:

$$FOM_{AMP} = \left( \frac{PSRR}{PSRR_{ref}} \cdot \frac{CMRR}{CMRR_{ref}} \cdot \frac{Gain}{Gain_{ref}} \cdot \frac{FOMS}{FOMS_{ref}} \cdot \frac{FOML}{FOML_{ref}} \right) \times \left( \frac{T_s}{T_{s_{ref}}} \cdot \frac{Area}{Area_{ref}} \right)^{-1} \times \left( \frac{v_n}{v_{n_{ref}}} \cdot \mathbb{I}(v_n > v_{n_{ref}}) \cdot \frac{TC}{TC_{ref}} \cdot \mathbb{I}(TC > TC_{ref}) \cdot \frac{vos}{vos_{ref}} \cdot \mathbb{I}(vos > vos_{ref}) \right)^{-1}, \quad (7)$$

$$FOM_S = \frac{GBW[MHz] \cdot C_{Load}[pF]}{Power[mW]}, \quad FOM_L = \frac{SR[V/\mu s] \cdot C_{Load}[pF]}{Power[mW]} \quad (8)$$

where  $v_n$  is the integrated noise,  $T_s$  is the settling time, and  $vos$  is the input offset voltage. PSRR represents the power supply rejection ratio, while CMRR denotes the common mode rejection ratio. TC is the temperature coefficient, GBW is the gain-bandwidth product,  $C_{Load}$  is the load capacitance.  $\mathbb{I}(\cdot)$  is an indicator function that equals 1 if the condition inside the parentheses is true and 0 otherwise.

Our experimental testing circuits include three multi-stage amplifiers (AMPs) from AnalogGym: NMCf, NMCNR, and DFCFC1, **with 11 optimization objectives and 5 constraints**. All experiments are conducted on a Linux workstation with 2 NVIDIA GeForce RTX 4080 GPUs and 4 Intel Xeon Platinum 8260 CPUs.

### A. Evaluating Accuracy of Transistor Sizing Mapping

To evaluate the accuracy of the transistor sizing mapping method in Section III-B, we compared the mapping and simulation results across three different processes: SMIC 180 nm, SKY 130 nm [21], and TSMC 90 nm. We used data from the BCOA, selecting 1,000 samples for each process.



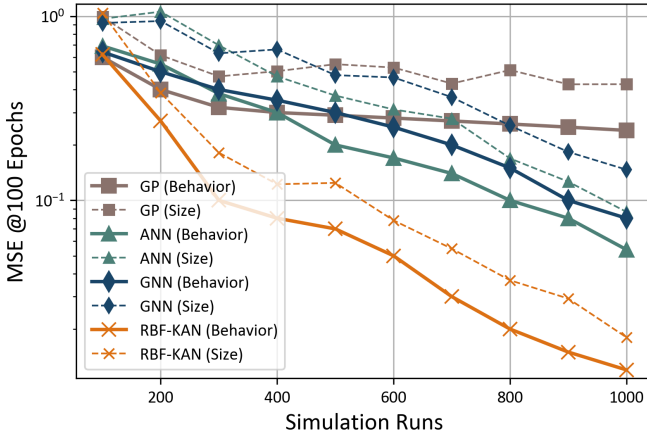


Fig. 6. Learning curves of average MSE over 10 runs for three AMPs.

Figure 5 shows that the mapping method achieves low errors, with  $g_m$  and  $f_t$  errors below 1% and  $V_{ds}$  and  $\Delta V_{gs}$  errors slightly higher but within 2%. The higher errors for  $V_{ds}$  and  $\Delta V_{gs}$  are due to their stronger sensitivity to device operating points and channel effects, which vary more significantly with process scaling. Overall, the mapping method maintains acceptable accuracy, demonstrating robustness for different process nodes.

#### B. Evaluating Nonlinear Fitting with Limited Samples

We conducted comparative experiments to evaluate the nonlinear fitting capability of our proposed RBF-KAN model. We used BagNet [22] to generate 10,000 samples. Then, by gradually increasing the training set, we evaluated the fitting capability of RBF-KAN under small-sample scenarios. Figure 6 shows mean squared error (MSE) at 100 epochs versus the number of simulation runs across different models: GP [6], ANN [2], GNN [23], and our RBF-KAN, tested for both behavior-based and transistor size-based modeling.

As shown in the figure, behavior-based modeling consistently achieves higher accuracy than size-based modeling. This is attributed to its decoupling of behavior from process dependence, thereby minimizing potential underfitting to the PDK. Compared to other models, RBF-KAN consistently achieves the highest accuracy. This is due to the Gauss-Laplace RBFs, which blend global smoothness and localized behavior capture. KAN's learnable activation functions also enable efficient handling of long-range feature interactions, even with minimal nodes. In many objective scenarios, GP requires covariance design to capture objective correlations, which compromises accuracy and extends processing times to over 10 hours due to joint modeling demands.

#### C. Comparison of BCOA and Other Methods

To demonstrate the effectiveness of the proposed BCOA framework, we compare  $FOM_{AMP}$  of BCOA with three state-of-the-art circuit sizing algorithms, including reinforcement learning based on relational GNN (rGNNRL) [5], constrained Voronoi tree search Bayesian optimization (cVTSBO) [15], and RL based on  $g_m/I_D$  ( $g_m/I_D$  RL) [24].

As shown in Table II, the improvements of BCOA primarily stem from significant enhancements in  $FOM_L$ ,  $FOM_S$ , area, and  $T_s$ . Other methods utilize the single-objective optimization paradigm, treating the transient response and area as constraints. However,

TABLE III  
COMPARISON OF HV AND MC-HV METHODS WITHIN THE BCOA

| Topology    | Processing time (min) ↓ |                  |                  | $FOM_{AMP}$ ↑       |                     |                   |
|-------------|-------------------------|------------------|------------------|---------------------|---------------------|-------------------|
|             | NMCF                    | NMCNR            | DFCFC            | NMCF                | NMCNR               | DFCFC             |
| HV          | 226 ± 11                | 313 ± 23         | 241 ± 19         | 182.4 ± 42.1        | 137.9 ± 37.7        | 3952 ± 884        |
| MC-HV       | <b>6.1 ± 0.9</b>        | <b>7.4 ± 1.2</b> | <b>6.8 ± 1.4</b> | <b>283.5 ± 21.4</b> | <b>186.2 ± 17.2</b> | <b>5863 ± 184</b> |
| Improvement | 97.3%                   | 97.6%            | 97.2%            | 1.55 ×              | 1.35 ×              | 1.48 ×            |

the sparse and narrow solution space for stable transient response, coupled with the lack of quantifiable feedback from oscillations, often leads to inefficient exploration and suboptimal solutions. Our framework employs the MC-HV method for efficient PF evaluation, dynamically adjusting reference points to guide many-objective optimization. As a result, BCOA significantly improves  $FOM_L$  by **1.76×** to **2.64×**,  $FOM_S$  by **1.73×** to **2.48×**,  $T_s$  by **1.31×** to **27.33×**, and reduces area by **15%-30%**. These individual improvements collectively contribute to the overall enhancement of  $FOM_{AMP}$ , which achieves gains ranging from **5.74×** to **98.05×** over  $g_m/I_D$  RL, **9.95×** to **176.60×** over cVTS, and **14.84×** to **184.37×** over rGNNRL.

As shown in Figure 7, While cVTSBO uses domain decomposition to accelerate convergence with fewer function evaluations (FEs), it often gets trapped in local optima under complete specifications, limiting further improvement after initial convergence. We also observe that other methods significantly improve in  $FOM_{AMP}$  after integrating behavior-level optimization. This is due to behavior-level representation, which establishes a smoother decision space, enabling more efficient exploration.

As shown in Table III, the MC-HV method significantly improves runtime and optimization guidance over the HV method. Using Monte Carlo-based sampling for the PF estimation, MC-HV reduces complexity in many-objective scenarios, decreasing runtime by about **97%**. Adaptive reference point adjustments and targeted high-curvature sampling allow MC-HV to concentrate on suboptimal regions, enhancing convergence and solution diversity, with an average  $FOM_{AMP}$  improvement of **1.35×** to **1.55×** across the three topologies.

The effectiveness and efficiency of BCOA are attributed to two factors. 1) We decouple the analog block from the process using behavior-level representation and employ RBF-KAN to build accurate, comprehensive surrogate models with limited simulations. 2) Using the MC-HV method to approximate the Pareto front, we guide multi-objective trade-offs, enabling efficient global optimization.

#### D. Design Porting among Process Nodes

To validate the efficiency and effectiveness of behavior-level representation in design porting, we conducted experiments across 180nm, 130nm, and 90nm process nodes. Specifically, we compared BCOA with knowledge transfer-based methods, such as GCNRL [4], to evaluate  $FOM_{AMP}$  and FE consumption across nodes. In GCNRL, technology transfer is accomplished by training an RL agent to size circuits in a simpler node and then applying the same agent to new nodes.

We use process-independent device electrical properties to represent circuit behavior, enabling optimized designs to be reused across nodes. Specifically, design porting is achieved by mapping optimized behavior to transistor sizes using the LUT of the new process. Furthermore, the behavior-based surrogate model captures general electrical characteristics, enabling efficient knowledge transfer. As shown in Figure 8, BCOA reduces FE consumption by **6.6×** to **7.8×** compared to size-based knowledge transfer in GCNRL, achieving over **16.2 ×** improvement in  $FOM_{AMP}$  across multiple nodes.

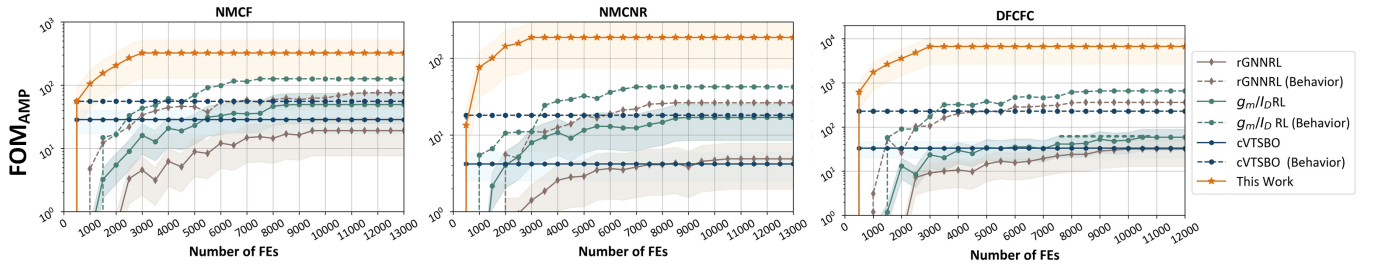


Fig. 7. Learning curves of different algorithms. The proposed BCOA consistently achieves the highest  $FOM_{AMP}$ .

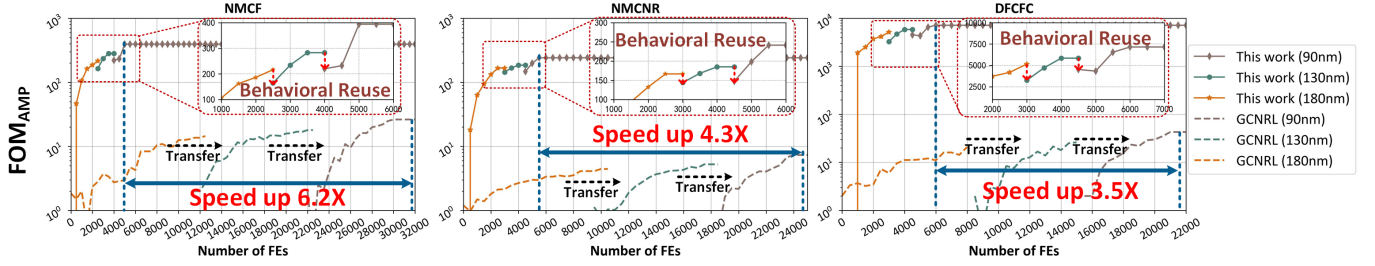


Fig. 8. Design porting among process nodes on AMPs. Behavioral representation enables cross-process reuse, while size-based representation demands substantial re-optimization at new node.

## V. CONCLUSION

We present BCOA, a behavior-level optimization framework for analog blocks. We decouple the design from specific processes by representing circuit behavior with transistor electrical properties rather than size. This enables accurate surrogate modeling with RBF-KAN under limited simulations and facilitates global optimization. We also employ the MCHV method for rapid PF evaluation across complete specifications. Extensive experiments show that BCOA outperforms other methods in  $FOM_{AMP}$  and provides strong knowledge transfer ability. Therefore, BCOA offers a more effective and efficient approach to analog block optimization and design porting.

## ACKNOWLEDGMENT

The project was supported by the National Key Research and Development Project of China (Grant No. 2023YFB4503002) and the Natural Science Foundation of China (Research Fund for International Senior Scientists, Grant No. W2431044).

## REFERENCES

- [1] S. Maji, A. F. Budak, S. Poddar, and D. Z. Pan, "Toward end-to-end analog design automation with ml and data-driven approaches (invited paper)," in *Proc. ASPDAC*, 2024, pp. 657–664.
- [2] A. F. Budak, M. Gandara, W. Shi, D. Z. Pan, N. Sun, and B. Liu, "An efficient analog circuit sizing method based on machine learning assisted global optimization," *IEEE TCAD*, vol. 41, no. 5, pp. 1209–1221, 2022.
- [3] J. Li, H. Zhi, W. Shan, Y. Li, Y. Zeng, and Y. Li, "Multi-task evolutionary to PVT knowledge transfer for analog integrated circuit optimization," in *Proc. ICCAD*, 2023.
- [4] H. Wang, K. Wang, J. Yang, L. Shen, N. Sun, H.-S. Lee, and S. Han, "GCN-RL circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning," in *Proc. DAC*, 2020, pp. 1–6.
- [5] Z. Li and A. C. Carusone, "Design and optimization of low-dropout voltage regulator using relational graph neural network and reinforcement learning in open-source sky130 process," in *Proc. ICCAD*, 2023, pp. 01–09.
- [6] W. Lyu, P. Xue, F. Yang, C. Yan, Z. Hong, X. Zeng, and D. Zhou, "An efficient bayesian optimization approach for automated optimization of analog circuits," *IEEE TCSI*, vol. 65, no. 6, pp. 1954–1967, 2018.
- [7] J. Li, Y. Zeng, H. Zhi, J. Yang, W. Shan, Y. Li, and Y. Li, "Knowledge transfer framework for pvt robustness in analog integrated circuits," *IEEE TCSI*, vol. 71, no. 5, pp. 2017–2030, 2024.
- [8] K. Hakhamaneshi, M. Nassar, M. Phielipp, P. Abbeel, and V. Stojanovic, "Pretraining graph neural networks for few-shot analog circuit modeling and design," *IEEE TCAD*, vol. 42, no. 7, pp. 2163–2173, 2023.
- [9] P. Jespers, *The  $g_m/I_D$  Methodology, a sizing tool for low-voltage analog CMOS Circuits*, 2009.
- [10] G. Shi, "Graph-pair decision diagram construction for topological symbolic circuit analysis," *IEEE TCAD*, vol. 32, no. 2, pp. 275–288, 2013.
- [11] T. Gu, W. Li, A. Zhao, Z. Bi, X. Li, F. Yang, C. Yan, W. Hu, D. Zhou, T. Cui, X. Liu, Z. Zhang, and X. Zeng, "BBGP-sDFO: Batch bayesian and gaussian process enhanced subspace derivative free optimization for high-dimensional analog circuit synthesis," *IEEE TCAD*, vol. 43, no. 2, pp. 417–430, 2024.
- [12] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, "KAN: Kolmogorov-arnold networks," *arXiv preprint arXiv:2404.19756*, 2024.
- [13] P. G. Jespers and B. Murmann, *Systematic design of analog CMOS circuits*. Cambridge University Press, 2017.
- [14] T. Gu, J. Wang, Z. Bi, C. Yan, F. Yang, Y. Qin, T. Cui, and X. Zeng, "tSS-BO: Scalable bayesian optimization for analog circuit sizing via truncated subspace sampling," in *Proc. DATE*, 2024, pp. 1–6.
- [15] A. Zhao, X. Wang, Z. Lin, Z. Bi, X. Li, C. Yan, F. Yang, L. Shang, D. Zhou, and X. Zeng, "cvts: A constrained voronoi tree search method for high dimensional analog circuit synthesis," in *Proc. DAC*, 2023, pp. 1–6.
- [16] J. Li, H. Zhi, R. Lyu, W. Li, Z. Bi, K. Zhu, Y. Zeng, W. Shan, C. Yan, F. Yang, Y. Li, and X. Zeng, "Analoggym: An open and practical testing suite for analog circuit synthesis," 2024. [Online]. Available: <https://arxiv.org/abs/2409.08534>
- [17] M. Stein, "Large sample properties of simulations using latin hypercube sampling," *Technometrics*, vol. 29, no. 2, pp. 143–151, 1987.
- [18] B. Liu, F. V. Fernández, Q. Zhang, M. Pak, S. Sipahi, and G. Gielen, "An enhanced moea/d-de and its application to multiobjective analog cell sizing," in *Proc. CEC*, 2010, pp. 1–7.
- [19] H. Wendland, *Scattered data approximation*. Cambridge university press, 2004, vol. 17.

- [20] J. Baxter, "A bayesian/information theoretic model of learning to learn via multiple task sampling," *Machine learning*, vol. 28, pp. 7–39, 1997.
- [21] "google/skywater-pdk," Aug. 2024, original-date: 2020-05-06T23:06:58Z. [Online]. Available: <https://github.com/google/skywater-pdk>
- [22] K. Hakhamaneshi, N. Werblun, P. Abbeel, and V. Stojanović, "Late breaking results: Analog circuit generator based on deep neural network enhanced combinatorial optimization," in *Proc. DAC*, 2019, pp. 1–2.
- [23] K. Hakhamaneshi, M. Nassar, M. Phielipp, P. Abbeel, and V. Stojanovic, "Pretraining graph neural networks for few-shot analog circuit modeling and design," *IEEE TCAD*, vol. 42, no. 7, pp. 2163–2173, 2023.
- [24] M. Choi, Y. Choi, K. Lee, and S. Kang, "Reinforcement learning-based analog circuit optimizer using gm/id for sizing," in *Proc. DAC*, 2023, pp. 1–6.